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Fachbereich Mathematik & Informatik

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Probabilistic Combinatorics SS24

Sheet 11

Due: 1. July at 19:00

(1) Consider the Erdös-Rényi model $G(n,p)$. Using the Janson-inequality show that the probability that a graph is triangle free is bounded from above by $e^{-\mu+\Delta/2}$ for $\mu = \binom{n}{3}p^3$ and $\Delta = 3\binom{n}{3}(n-3)p^5$

(2) Consider the Erdös-Rényi model $G(n,p)$ of random graphs.

Using the results about correlation inequalities show that

$$P\big(\, G(n,p) \text{ is triangle free}\,\big) \geq (1-p^3)^{\binom{n}{3}}$$

(3) Let $B_1, ..., B_r$, $\mathbb{P}$, X , μ, and Δ be like in the setup of the Janson-inequality. For every $T \subset [r]$, let

$$\begin{aligned} X_T &= \sum_{i\in T} X_i \\ \mu_T &= \mathbb{E}(X_T) \\ \Delta_T &= \sum_{\substack{i,j\in T, i\neq j \\ A_i\cap A_j \neq \emptyset}} \mathbb{P}(B_i \cap B_j) \end{aligned}$$

(i) Show that for every $T \subset [r]$ it holds, that

$$\mathbb{P}(X=0) \leq e^{-\mu_T + \frac{\Delta_T}{2}}$$

(ii) Now choose T randomly by independently including every $i \in [r]$ with probability q. Prove that:

$$\mathbb{E}(-\mu_T + \frac{\Delta_T}{2}) = -q\mu + \frac{q^2\Delta}{2}$$

(4) (Second Janson Inequality) Consider the setup of (3), but now also assume that $\Delta \geq \mu$. Use (3) to prove that:

$$\mathbb{P}(X=0) \leq e^{-\frac{\mu^2}{2\Delta}}$$

Hint: Choose q to minimize $\mathbb{E}(-\mu_T + \frac{\Delta_T}{2})$.